MIL-B-52589C(ME)
30 March 1987
SUPERSEDING
MIL-B-52589B(ME)
7 May 1980

MILITARY SPECIFICATION

BOOM, CENTERPOLE, BOLTED TANK ERECTION

This specification is approved for use within the USA Belvoir Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a 9-foot center pole boom for bolted tank erection.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

FF-B-575	- Bolts, Hexagon and Square.
FF-N-836	- Nut: Square, Hexagon Cap, Slotted, Castle Knurled, Welding and Single Ball Seat.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3950
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AMSC N/A
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QQ-A-200/8	- Aluminum Alloy Wire, Rod, Bar, Shapes and Tube Extruded, 6061.
QQ-A-225/8	- Aluminum Alloy Wire, Rod and Bar; Rolled or Cold Finished, 6061.
QQ-A-250/11	- Aluminum Alloy 6061, Plate and Sheet.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-P-25995	- Pipe, Aluminum Alloy, Drawn or Extruded.
MIL-C-46168	- Coating, Aliphatic, Polyurethane, Chemical Agent Resistant.

STANDARDS

FEDERAL

FED-STD-H28	- Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-889	- Dissimiliar Metals.

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the non-Government documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practices for Commercial Packaging.

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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AMERICAN WELDING SOCIETY, INC. (AWS)

AWS B2.1 - Welding Procedure and Performance Qualification.
ANSI/AWS D1.2 - Structural and Welding Code - Aluminum

(Application for copies should be addressed to American Welding Society, Inc., 550 NW Lejeune Road, P.O. Box 351040, Miami, FL 33135)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specifications sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The center pole boom, hereinafter referred to as "boom", shall be as shown in figure 1 and as specified herein, with a maximum capacity of 500 pounds.

3.2 Materials. Materials shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 3.2.6).

3.2.1 Pipe. Aluminum pipe shall conform to MIL-P-25995, type I, 6061-T6, schedule 10.

3.2.2 Plate. Aluminum plate shall conform to QQ-A-250/11, 6061-T6.

3.2.3 Rod. Aluminum rod shall conform to QQ-A-200/8 or QQ-A-225/8, 6061-T6.

3.2.4 Bolts. Steel bolts shall conform to FF-B-575, type I, size as shown in figure 1.

3.2.5 Nuts. Steel nuts shall conform to FF-N-836, type I, style 1.

3.2.6 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the booms may be newly fabricated from recovered materials to the maximum extent practicable, provided the booms produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the booms.

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3.2.7 Material deterioration and control. The booms shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operation and storage environments to which the item may be exposed.

3.2.8 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.2.9 Identification of material and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent and shall make information available upon request to the contracting officer or designated representative.

3.3 Marking.

3.3.1 Identification marking. The boom shall be identified in accordance with MIL-STD-130.

3.3.2 Working load. The boom capacity shall be stenciled in 2-inch letters "LOAD CAPACITY 500 LBS" in accordance with MIL-T-704.

3.4 Treatment and painting. The portions of the boom normally painted internally and externally should be cleaned, treated, and painted in accordance with MIL-T-704, type F or G, as applicable. Unless otherwise specified (see 6.2), top coat color shall be camouflage green 383 conforming to MIL-C-46168.

3.5 Workmanship.

3.5.1 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause damage to the material. Corners shall be square and true. Flame-cutting of steel, using tips suitable for the thickness of the metal, may be used instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Heated steel shall be allowed to cool slowly. Plasma arc or mechanical cutting shall be used for aluminum materials. Flame cutting of aluminum shall not be used. Precaution shall be taken to avoid overheating. All external surface shall be free of burrs, sharp edges and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

3.5.2 Welding. The aluminum surfaces of parts to be welded shall be free from paint, grease, oxide film, and other foreign matter. The welding shall be done by the inert gas, shielded-arc method. Welds may be ground, filed, buffed, or chipped but shall not be hammered, and shall be free from discontinuities such as base metal cracks adjacent to or behind welds, cracks in weld metal, undercutting of base metal, lack of fusion with parent metal or between parts, lack of penetration, lack of prescribed fit, or spatter.

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3.5.3 Welders. Before assigning any welder to welding work covered by this specification, the contractor shall obtain certification that the welder or welding operator has passed qualification tests as prescribed by either of the following listed codes for this type of welding operation to be performed and that such qualification is effective as defined by the particular code:

AWS B2.1 - Welding Procedure and Performance Qualification.
ANSI/AWS D1.2 - Structural Welding Code - Aluminum.

Contractors who only make flat or horizontal welds need not qualify welders for "all position welding". Subject to approval by the Government, the contractor's standard welder qualification may be substituted in lieu of the above codes provided that the contractor's procedure is equivalent to the above codes. The contractor shall be responsible for determing that automatic welding equipment operators are capable of producing quality welds in accordance with AWS, and ANSI/AWS codes. The certification shall be made available for review by the contracting officer or the contracting officer's representative.

3.5.4 Filler metal. Aluminum welding filler metal shall be alloy 4043 or 5356.

3.5.5 Bolted connections. Boltholes shall be accurately formed and shall have the burrs removed. Washers or lockwashers shall be provided where necessary. Matching thread areas securing bolts or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. All threads shall conform to FED-STD-H28. All fasteners shall have full thread engagement. Bolts shall protrude not more than 2 full threads.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Quality conformance inspection (see 4.3).
b. Inspection of packaging (see 4.5).

4.3 Quality conformance inspection.

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4.3.1 Lot. All booms offered for delivery at one time shall constitute a lot.

4.3.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.3.3 Examination. Samples selected in accordance with 4.3.2 shall be examined as specified in 4.4.1. Acceptable quality level (AQL) shall be 1.0 percent defective.

4.4 Inspection procedure.

4.4.1 Examination. The boom shall be examined as specified herein for the following defects:

Major

101. Boom height, boom length, or other dimensions not as specified.
102. Materials not as specified.
103. Used, rebuilt, or remanufactured components, pieces or parts incorporated into the boom.
104. Materials not resistant to corrosion or deterioration or treated to be made resistant to corrosion or deterioration for the applicable storage and operating environment.
105. Dissimilar metals, as defined in MIL-STD-889, not effectively insulated from each other.
106. Contractor does not have documentation available for identification of material, material finishes or treatment.
107. Marking not as specified.
108. Treatment and painting not as specified.
109. Workmanship not as specified.
110. Welders certificate not as specified.
111. Welding and welders not as specified.
112. Bolted connections not as specified.

Minor.

201. Identification marking missing, incomplete or illegible.

4.5 Inspection of packaging.

4.5.1 Quality conformance inspection of pack.

4.5.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.5.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.5.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the following defects. AQL shall be 1.0 percent defective.

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113. Boom not unassembled as specified for level A or commercial.
114. Bolts and pins not preserved as specified for level A.
115. Preservation and packing not in accordance with the referenced document specified for commercial.
116. Box not as specified for level A or B.
117. Contents not blocked, braced, or cushioned to prevent movement or damage as specfified for level A or B.
118. Marking missing, illegible, incorrect, or incomplete for level A, B, or commercial.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial, as specified (see 6.2).

5.1.1 Level A. The boom shall be unassembled. Loose bolts and pins shall be preserved together in accordance with MIL-P-116, method IC-1, applying type P-1 preservative if the items are unpainted.

5.1.2 Commercial. The boom shall be unassembled and preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, level B, or commercial, as specified (see 6.2).

5.2.1 Level A. The unassembled boom, preserved as specified in 5.1, shall be packed in a close-fitting box conforming to PPP-B-601, overseas type, style optional or PPP-B-621, class 2, style optional. Blocking, bracing, cushioning and immobilization of the contents within the container shall meet the requirements of MIL-P-116, including the applicable rough handling tests. Box closure and strapping shall be in accordance with the appendix to the box specification.

5.2.2 Level B. Packing shall be as specified in 5.2.1 for level A except the box shall be domestic type.

5.2.3 Commercial. Commercial packing shall be in accordance with ASTM D 3951.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection (level A or B) shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Marking for commercial packaging shall be in accordance with ASTM D 3951. In addition, each shipping container shall display weight and cube data.

6. NOTES

6.1 Intended use. The boom is used for erecting bolted steel storage tanks. When erection is complete, the pole and boom are permanently incorporated into the tank as support for the roof.

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6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Degree of preservation and degree of packing required (see 5.1 and 5.2).
c. Color required when other than as specified (see 3.4).

6.3 Subject term (key word) listing.

Bolted tank erection
Boom
Center pole

6.4 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

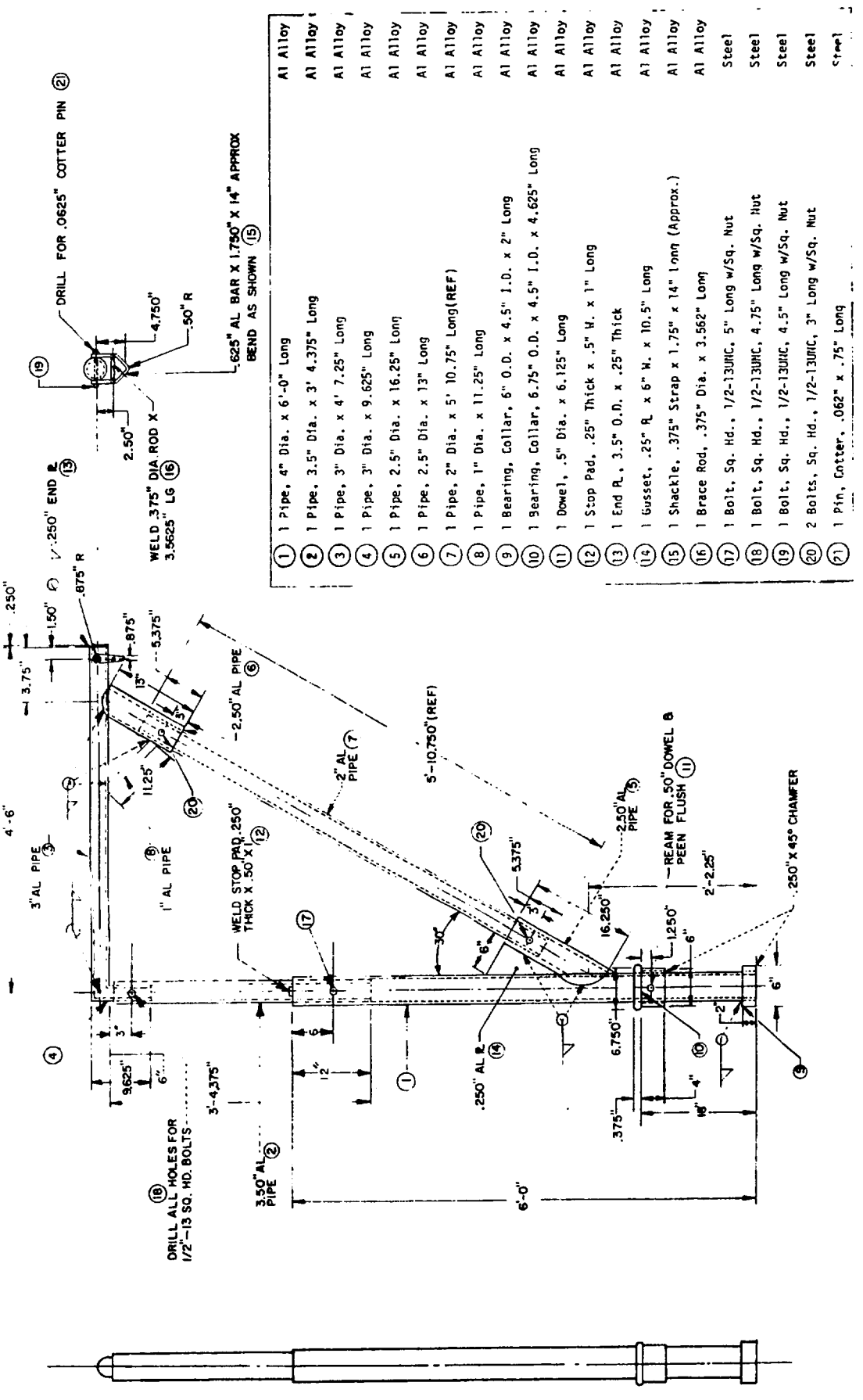
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(1)	1 Pipe, 4" Dia. x 6'-0" Long	Al Alloy
(2)	1 Pipe, 3.5" Dia. x 3' 4.375" Long	Al Alloy
(3)	1 Pipe, 3" Dia. x 4' 7.25" Long	Al Alloy
(4)	1 Pipe, 3" Dia. x 9.625" Long	Al Alloy
(5)	1 Pipe, 2.5" Dia. x 16.25" Long	Al Alloy
(6)	1 Pipe, 2.5" Dia. x 13" Long	Al Alloy
(7)	1 Pipe, 2" Dia. x 5' 10.75" Long(REF)	Al Alloy
(8)	1 Pipe, 1" Dia. x 11.25" Long	Al Alloy
(9)	1 Bearing, Collar, 6" O.D. x 4.5" I.D. x 2" Long	Al Alloy
(10)	1 Bearing, Collar, 6.75" O.D. x 4.5" I.D. x 4.625" Long	Al Alloy
(11)	1 Dowel, .5" Dia. x 6.125" Long	Al Alloy
(12)	1 Stop Pad, .25" Thick x .5" W. x 1" Long	Al Alloy
(13)	1 End PL, 3.5" O.D. x .25" Thick	Al Alloy
(14)	1 Gusset, .25" PL x 6" W. x 10.5" Long	Al Alloy
(15)	1 Shackle, .375" Strap x 1.75" x 14" Long (Approx.)	Al Alloy
(16)	1 Brace Rod, .375" Dia. x 3.562" Long	Al Alloy
(17)	1 Bolt, Sq. Hd., 1/2-13UNC, 5" Long w/Sq. Nut	Steel
(18)	1 Bolt, Sq. Hd., 1/2-13UNC, 4.75" Long w/Sq. Nut	Steel
(19)	1 Bolt, Sq. Hd., 1/2-13UNC, 4.5" Long w/Sq. Nut	Steel
(20)	2 Bolts, Sq. Hd., 1/2-13UNC, 3" Long w/Sq. Nut	Steel
(21)	1 Pin, Cotter, .062" x .75" Long	Steel

FIGURE 1. Boom, center pole bolted tank erection.

X-981 B

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-52589C(ME)	Boom, Centerpole, Bolted Tank Erection

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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